

Voice Conversion 500-Pair

Out-of-Distribution Evaluation Report

EE 123 Project — Ved, Ching, Erick

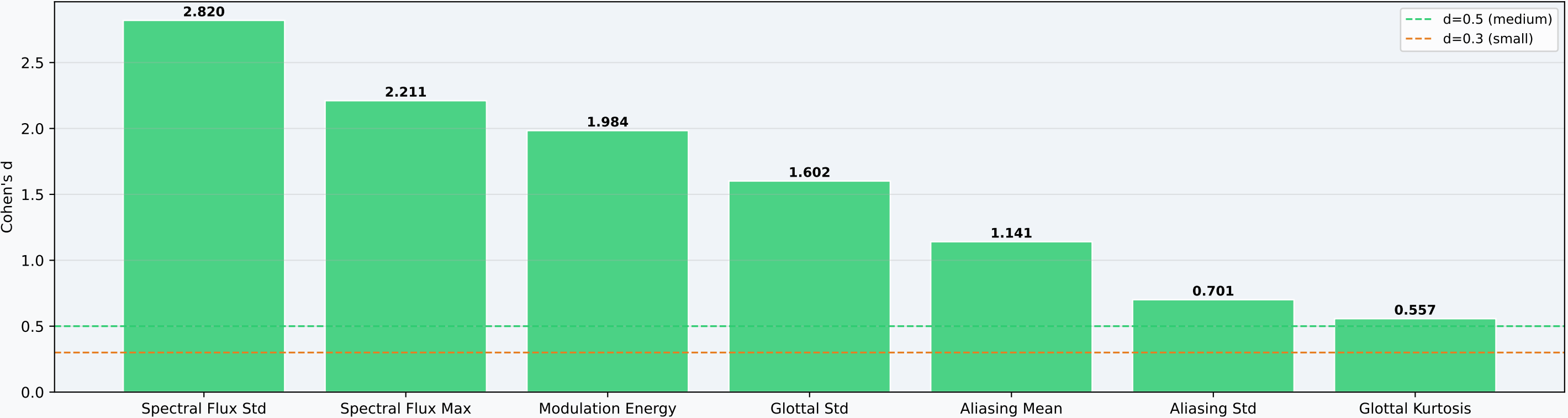
Real : 500 files (source-vctk500)

Spoof: 500 files (converted-vctk500)

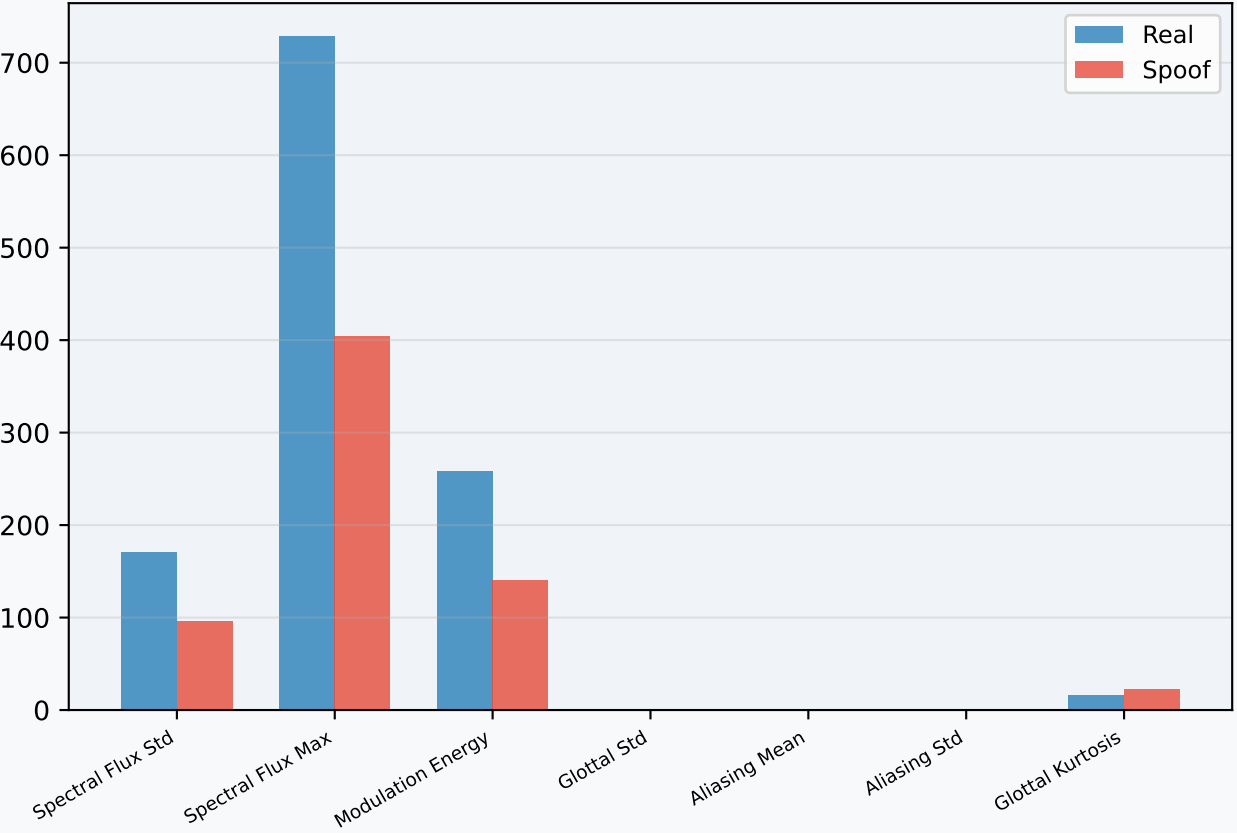
Models trained on VCTK 1000 vs 1000 (balanced)

Feature Discriminability — Voice Conversion 500 Pairs

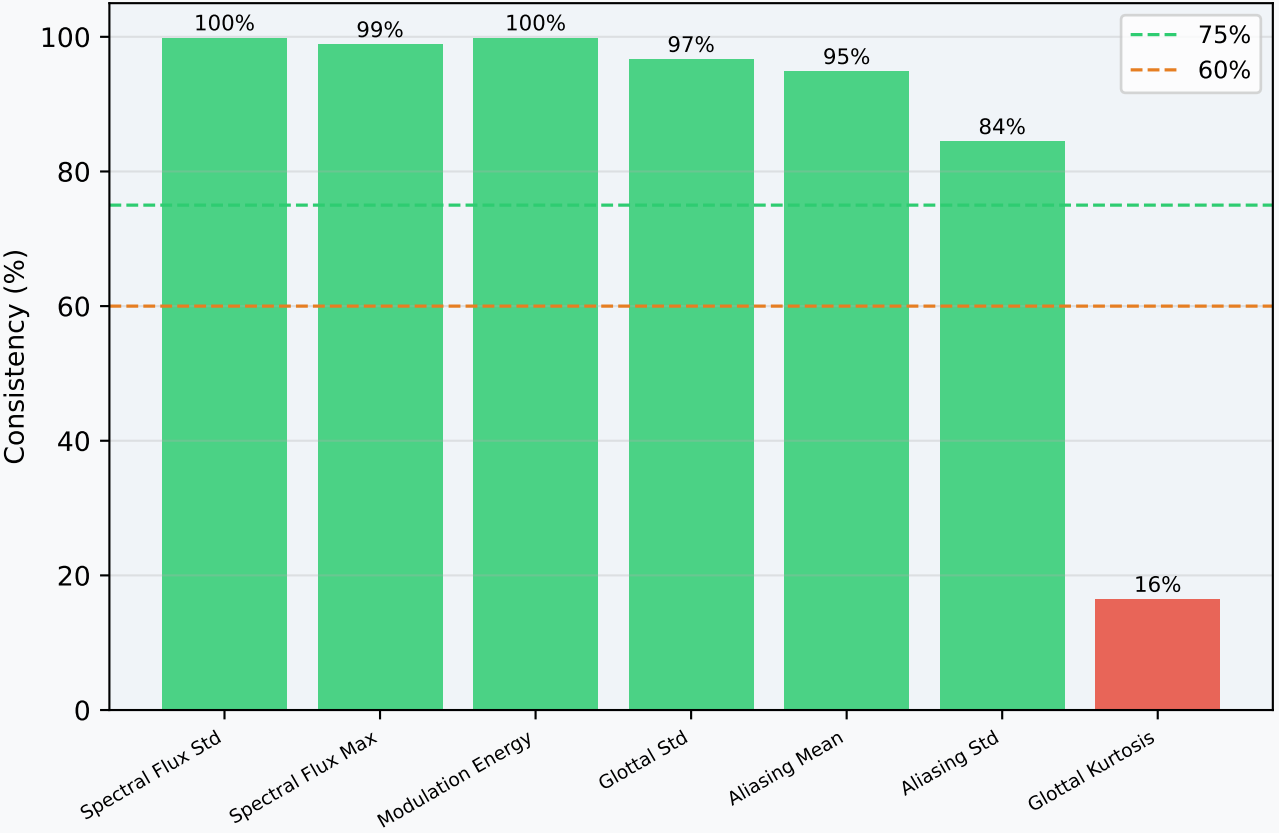
Cohen's d — Feature Discriminability



Mean Feature Values: Real vs Spoof



Consistency: % Pairs where Real > Spoof



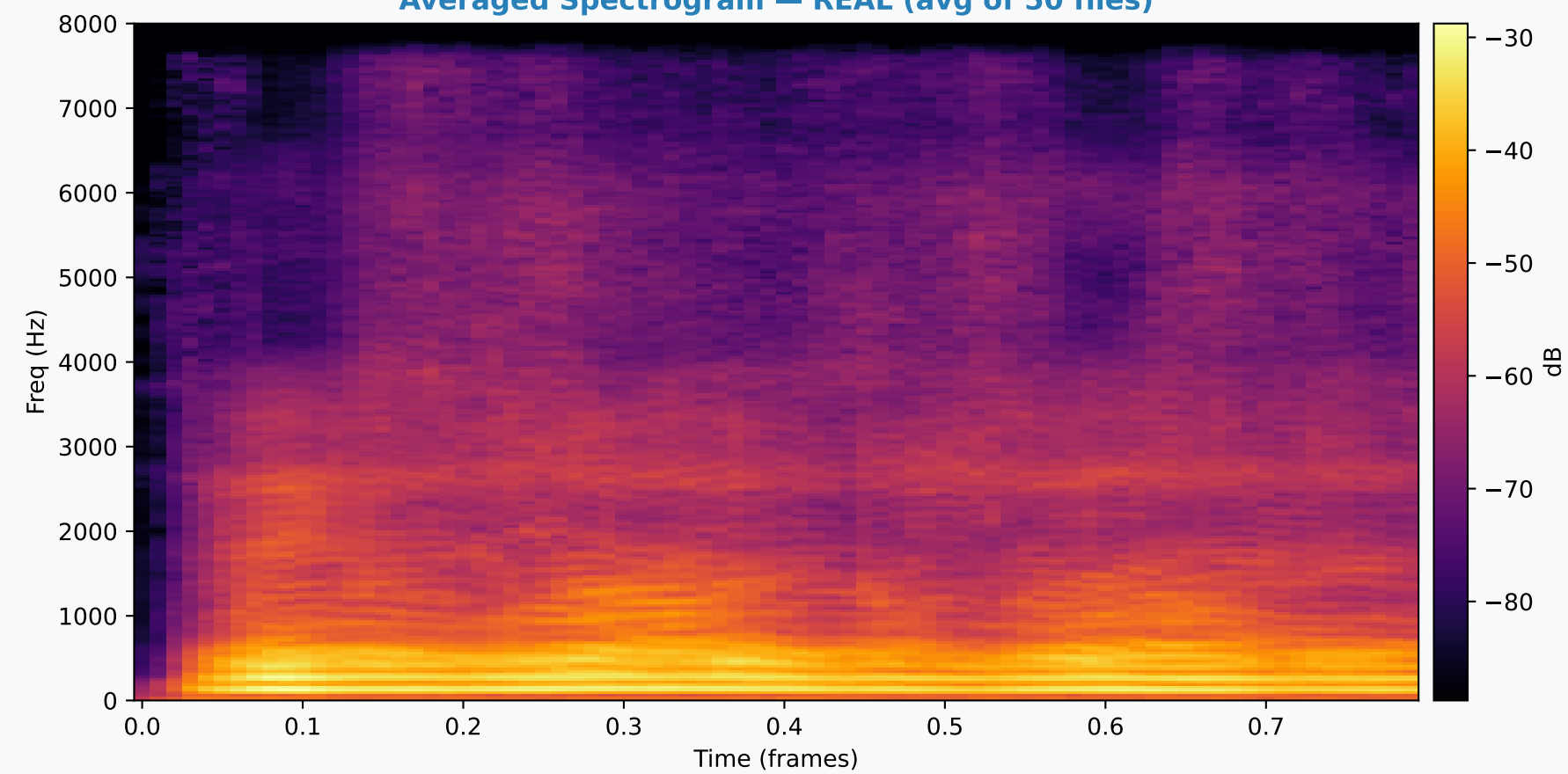
Voice Conversion 500-Pair Aggregate Feature Statistics

Feature	Real Mean	Real Std	Spoof Mean	Spoof Std	Diff%	Cohen's d	Real>Spoof (out of 500 pairs)	Consist%	Keep?
Spectral Flux Std	170.211	34.053	95.277	15.888	+44.0%	2.820	499/500	99.8%	✓ KEEP
Spectral Flux Max	727.999	186.407	404.259	90.178	+44.5%	2.211	494/500	98.8%	✓ KEEP
Modulation Energy	257.555	74.819	140.302	37.264	+45.5%	1.984	499/500	99.8%	✓ KEEP
Glottal Std	0.017	0.006	0.009	0.003	+45.3%	1.602	483/500	96.6%	✓ KEEP
Aliasing Mean	0.121	0.068	0.060	0.033	+50.4%	1.141	474/500	94.8%	✓ KEEP
Aliasing Std	0.160	0.106	0.101	0.056	+37.1%	0.701	422/500	84.4%	✓ KEEP
Glottal Kurtosis	15.956	8.793	22.337	13.602	-40.0%	0.557	82/500	16.4%	✗ DROP

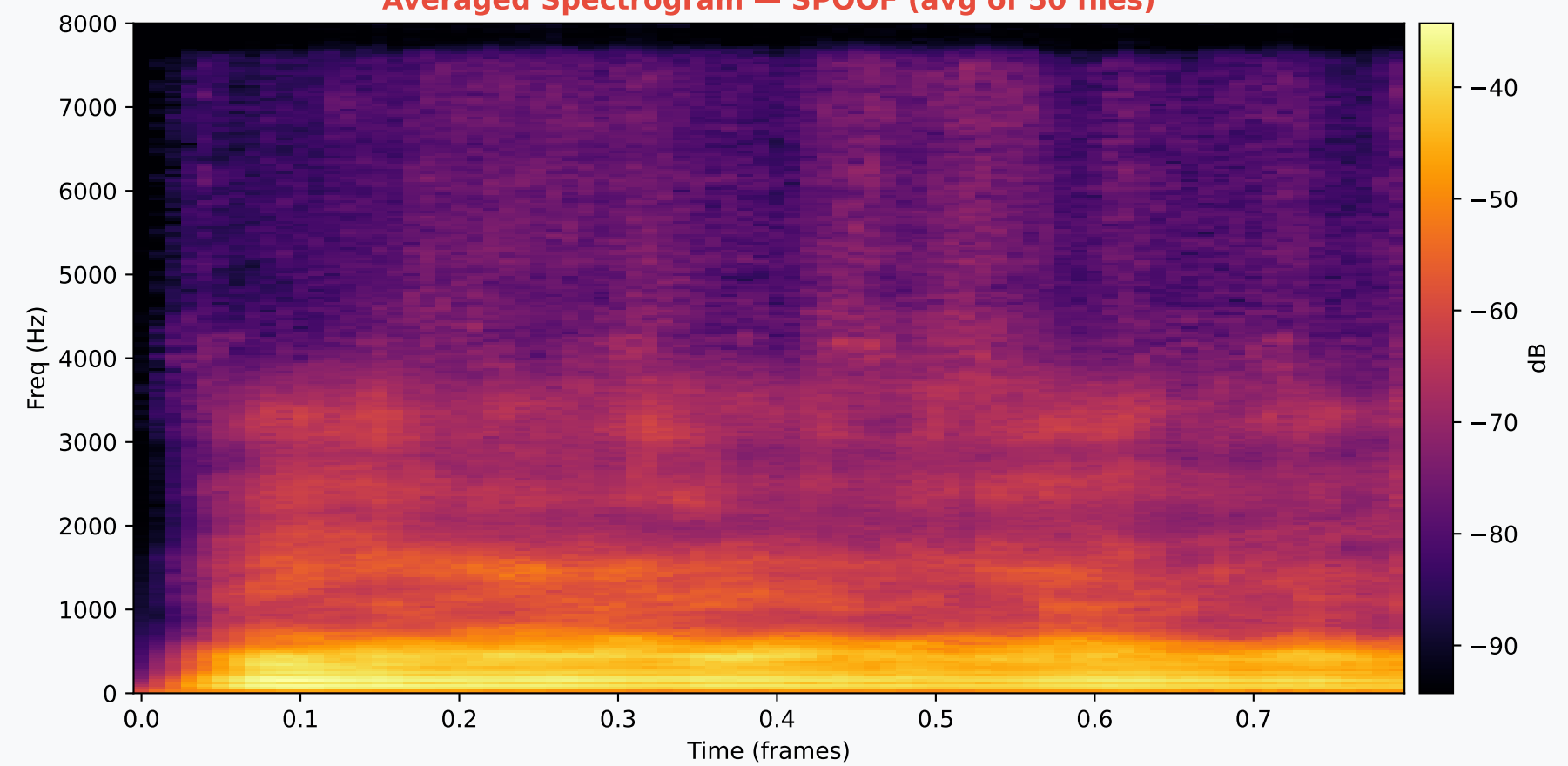
Green = strong (d>0.5) | Yellow = moderate (d>0.3) | Red = weak/drop

Averaged Signal Analysis — VC 500 Pairs (50 files sampled per class)

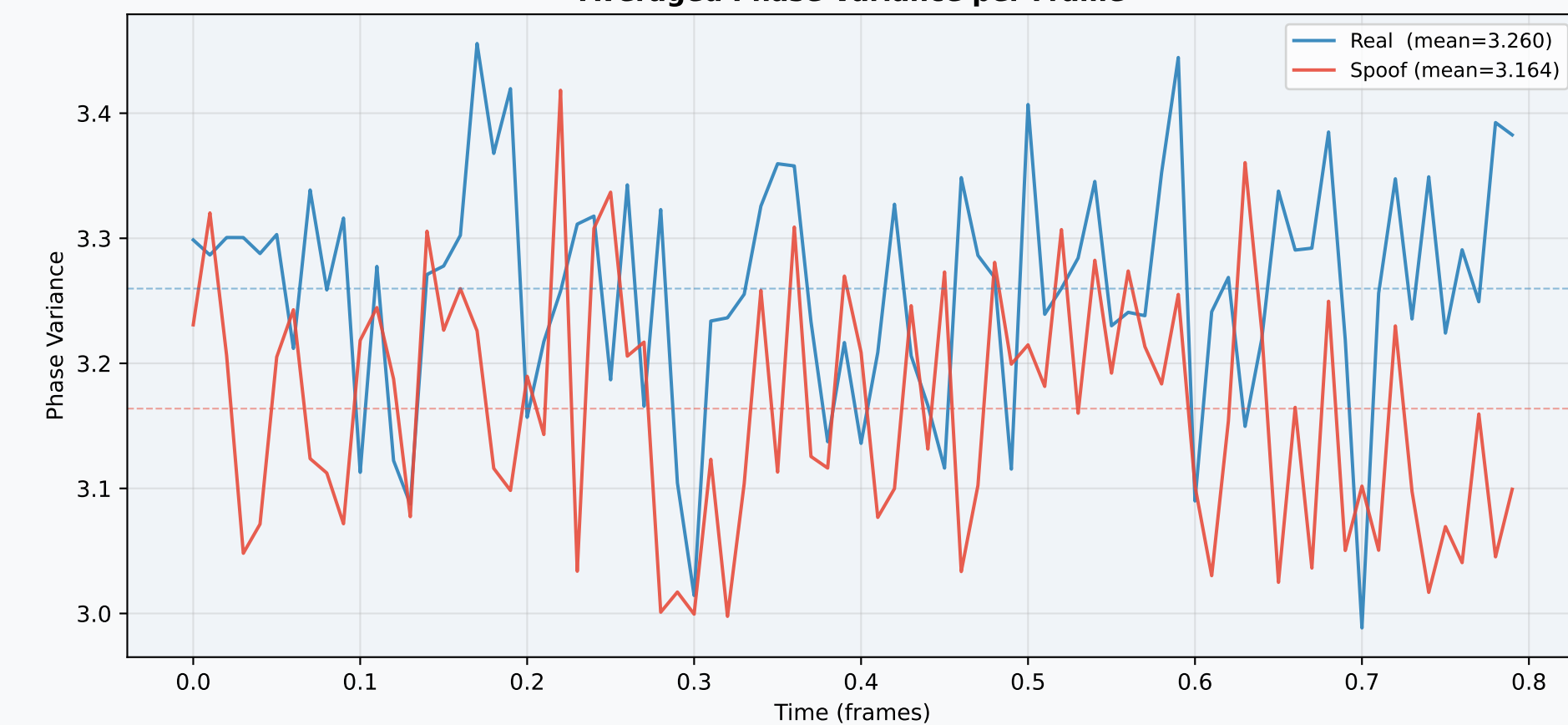
Averaged Spectrogram — REAL (avg of 50 files)



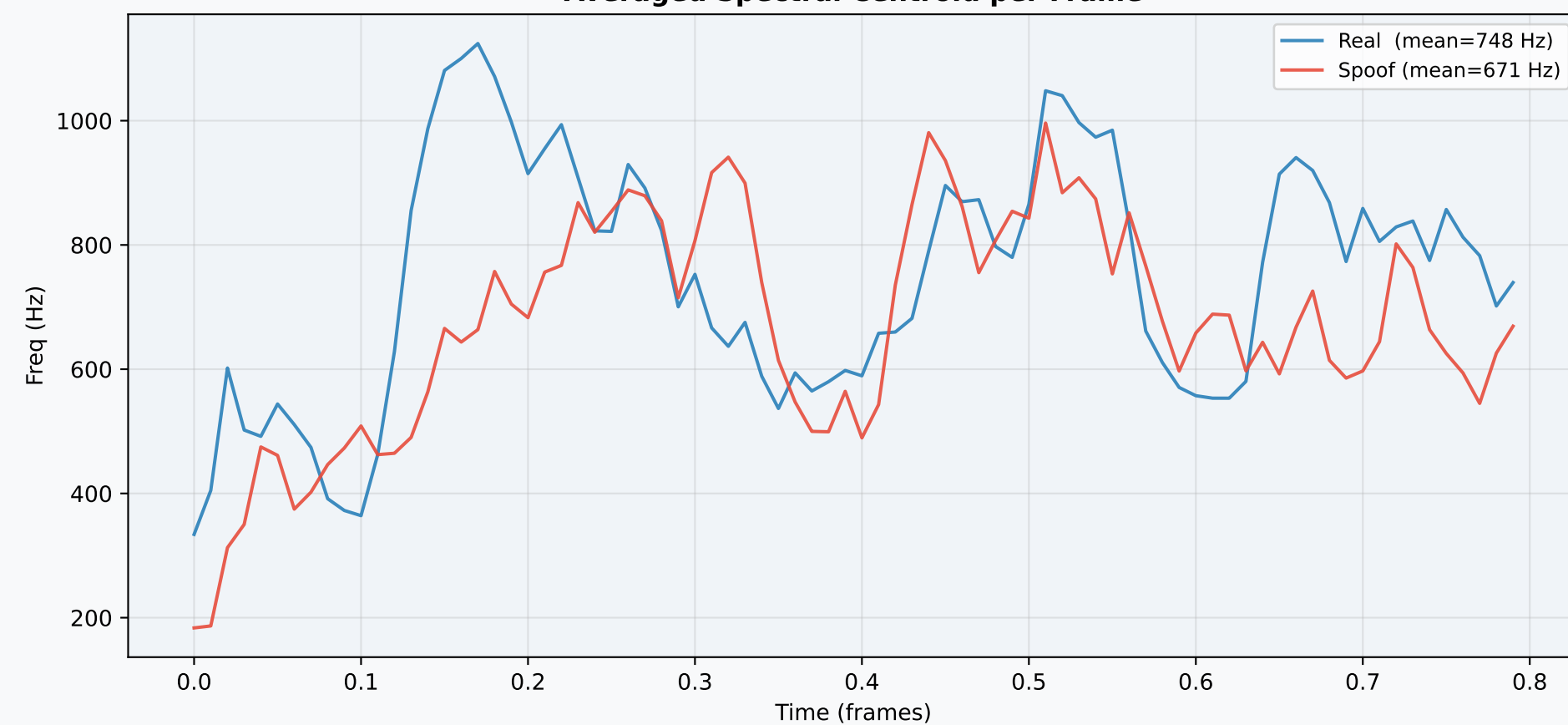
Averaged Spectrogram — SPOOF (avg of 50 files)



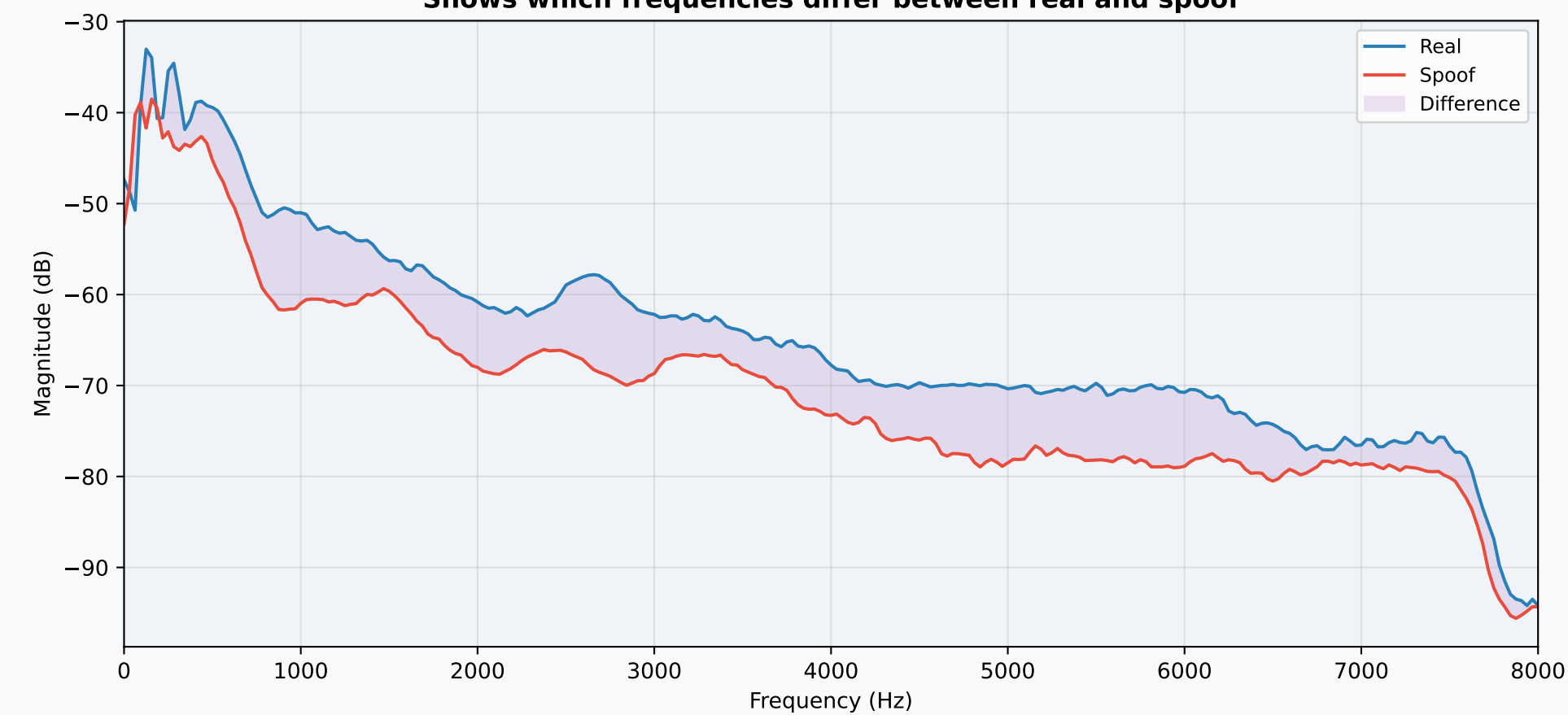
Averaged Phase Variance per Frame



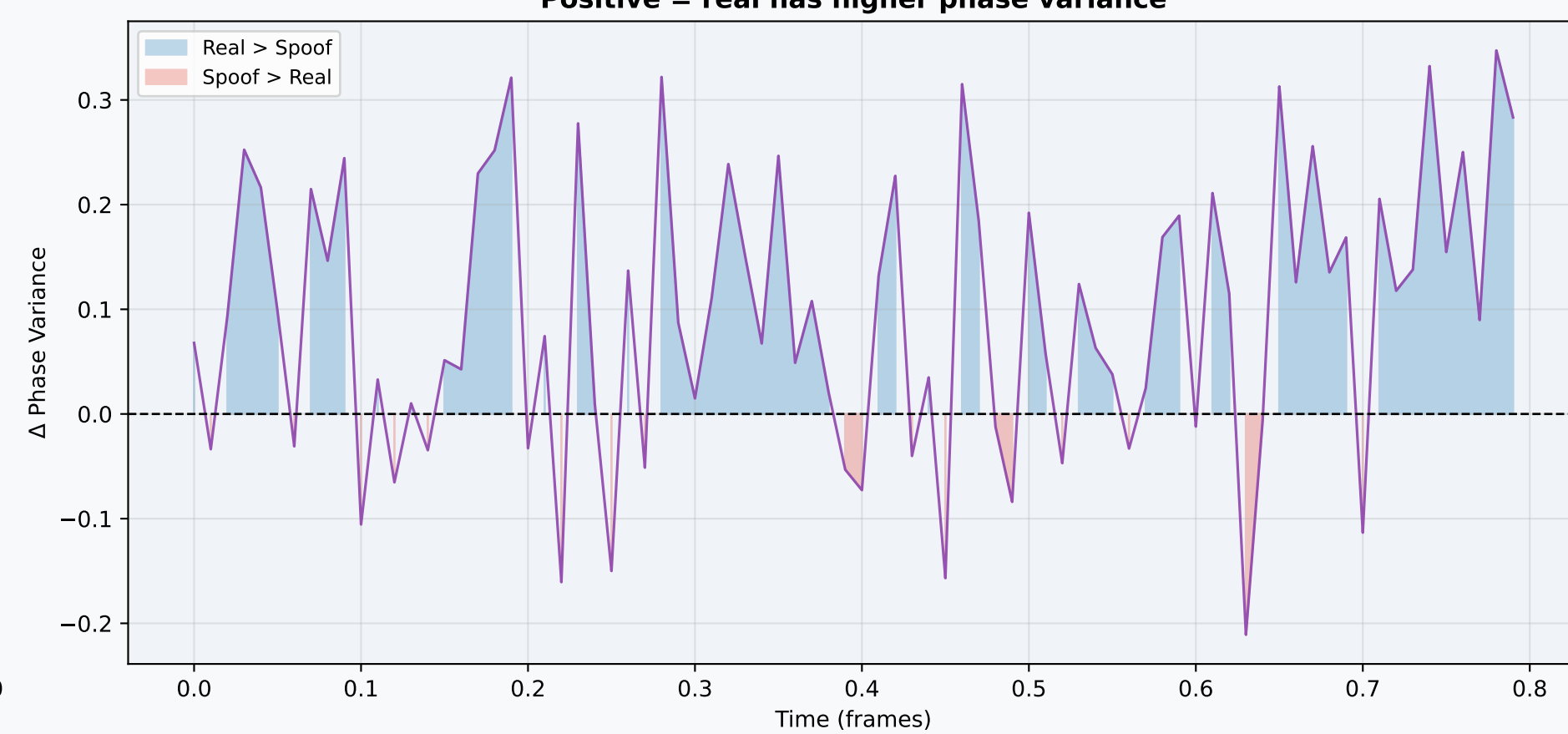
Averaged Spectral Centroid per Frame



Averaged Magnitude Spectrum (dB)
Shows which frequencies differ between real and spoof

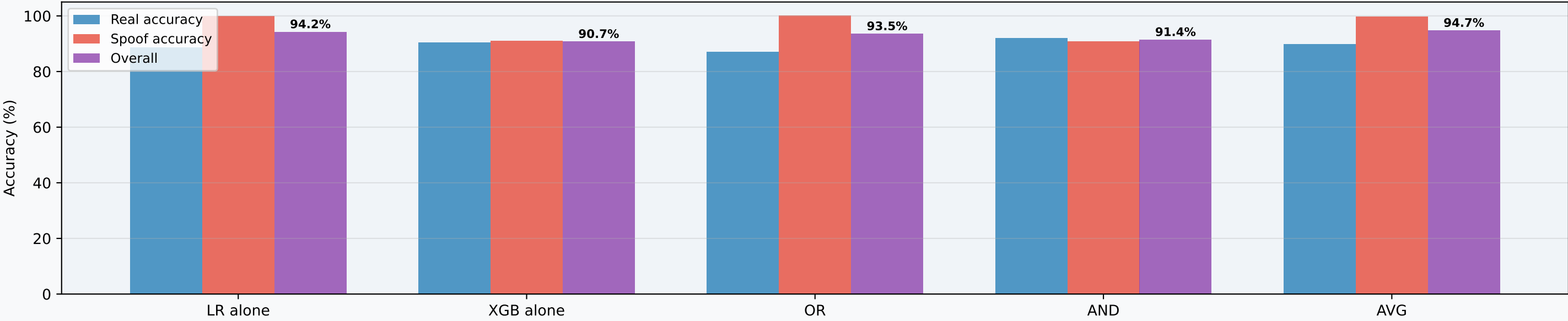


Phase Variance Difference (Real - Spoof)
Positive = real has higher phase variance

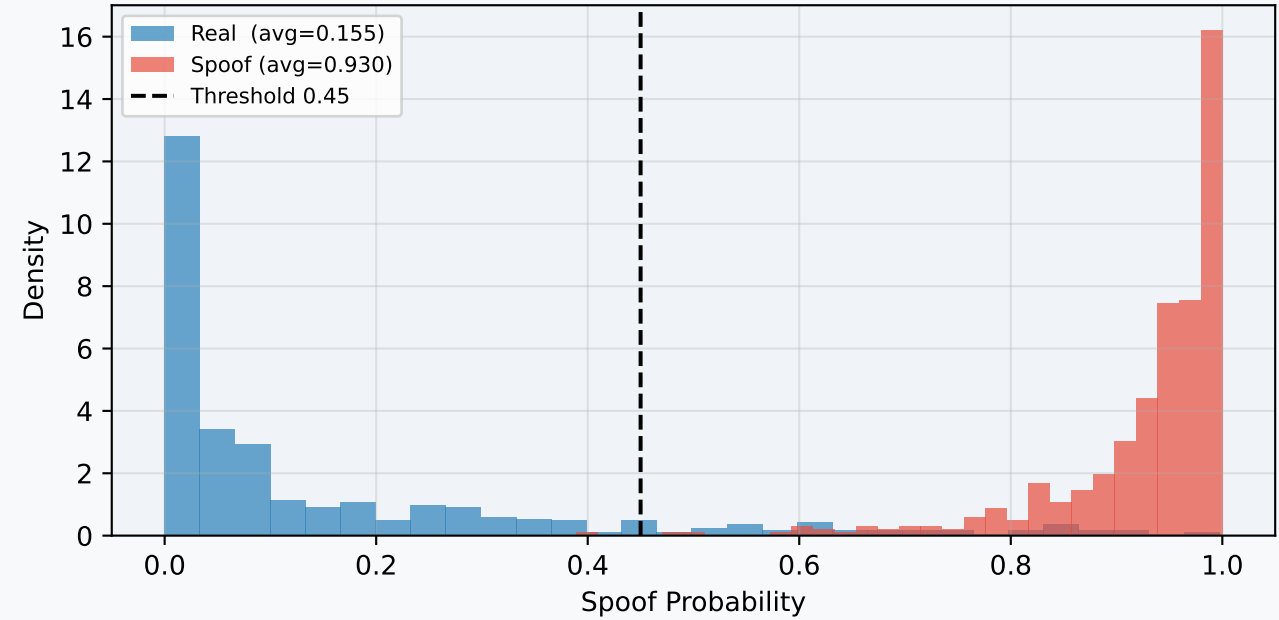


Voice Conversion 500-Pair — Ensemble Evaluation Results

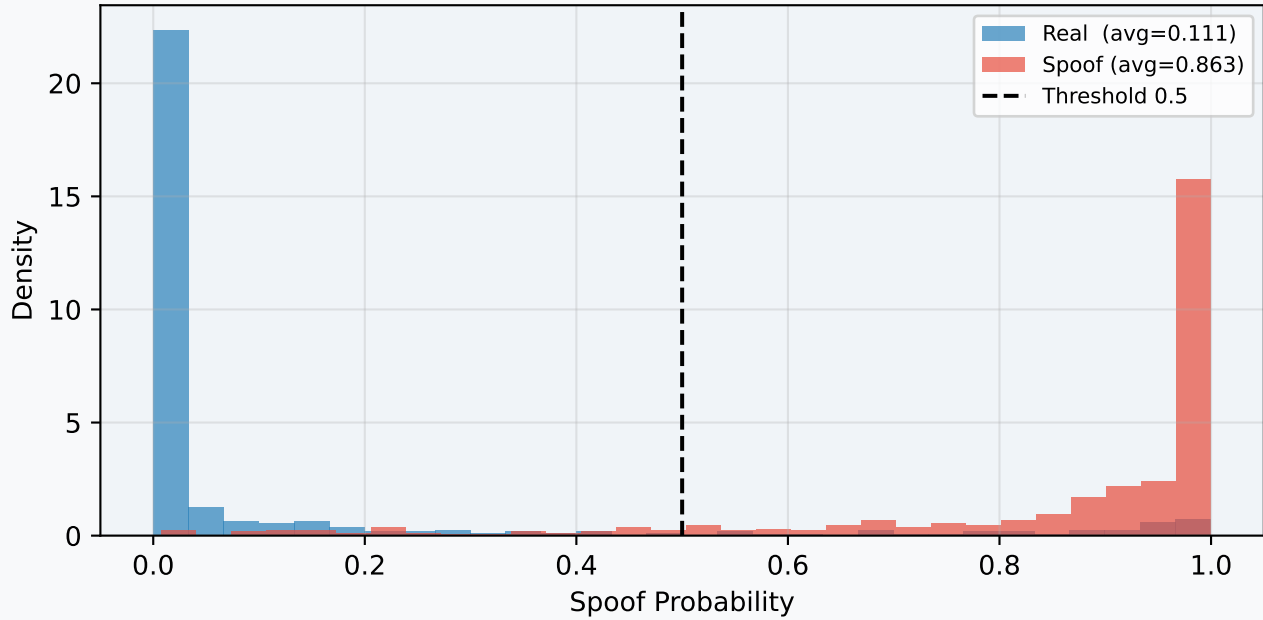
Accuracy by Strategy



LR Probability Distribution



XGBoost Probability Distribution



Raw Count Summary

Strategy	Real Correct	Real Total	Real%	Spoof Correct	Spoof Total	Spoof%	Overall Correct	Overall Total	Overall%
LR alone	443	500	88.6%	499	500	99.8%	942	1000	94.2%
XGB alone	452	500	90.4%	455	500	91.0%	907	1000	90.7%
OR	435	500	87.0%	500	500	100.0%	935	1000	93.5%
AND	460	500	92.0%	454	500	90.8%	914	1000	91.4%
AVG	449	500	89.8%	498	500	99.6%	947	1000	94.7%